

Binomial Coefficients in Expansion and Probability

Answer Key

Precalculus / Advanced Statistics Discrete Mathematics

Quick Answers

- | | |
|-------------------------------------|-------------------------------|
| 1. 35 | 7. 1512 |
| 2. $x^3 + 6x^2 + 12x + 8$ | 8. 10, 7, 220, $\frac{7}{22}$ |
| 3. 4, 84, $\frac{1}{21}$ | 9. $\frac{3087}{10000}$ |
| 4. 84, 126, 210 | 10. 160 |
| 5. $16x^4 - 32x^3 + 24x^2 - 8x + 1$ | 11. 10, $\frac{5}{16}$ |
| 6. 15, $\frac{15}{64}$ | 12. 28, $\frac{5103}{16384}$ |

Curriculum

Level: Precalculus / Advanced Statistics Discrete Mathematics

HSA-APR.C.5 – *problems 1, 2, 4, 5, 7, 10*

HSS-CP.A–HSS-CP.B – *problems 3, 6, 8, 9, 11, 12*

Difficulty 1–5 of 5 across 24 tagged checks.

1. Evaluate $\binom{7}{3}$.

Step 1. $\binom{7}{3} = \frac{7!}{3!4!}$. Cancel the 4! against the tail of 7! before multiplying anything out.

Step 2. $\frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = \frac{210}{6} = 35$.

Step 3. Sanity check with the symmetry $\binom{7}{3} = \binom{7}{4}$: choosing which 3 to take is the same as choosing which 4 to leave. 35

2. Expand $(x + 2)^3$.

Step 1. Row 3 of Pascal's triangle is 1, 3, 3, 1, so $(x + 2)^3 = x^3 + 3x^2(2) + 3x(2)^2 + (2)^3$.

Step 2. Each term needs its power of 2 as well as its coefficient: $3 \cdot 2 = 6$, $3 \cdot 4 = 12$, $2^3 = 8$.

Step 3. Collecting: $x^3 + 6x^2 + 12x + 8$. Check by substituting $x = 1$: the expansion gives 27, and $(1 + 2)^3 = 27$ ✓ $x^3 + 6x^2 + 12x + 8$

3. Committee of 3 from 9 members, 4 of them seniors.

Step 1 (part a). Order does not matter on a committee, so the total is $\binom{9}{3} = \frac{9 \cdot 8 \cdot 7}{3 \cdot 2 \cdot 1} = 84$. 84

Step 2 (part b). An all-senior committee uses 3 of the 4 seniors: $\binom{4}{3} = 4$. 4

Step 3 (part c). All 84 committees are equally likely, so the probability is $\frac{4}{84} = \frac{1}{21}$. $\frac{1}{21}$

4. Pascal's rule for $\binom{9}{3}$, $\binom{9}{4}$, $\binom{10}{4}$.

Step 1. $\binom{9}{3} = \frac{9 \cdot 8 \cdot 7}{6} = 84$ and $\binom{9}{4} = \frac{9 \cdot 8 \cdot 7 \cdot 6}{24} = 126$.

$$\boxed{84 \quad 126}$$

Step 2. $\binom{10}{4} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{24} = 210$.

$$\boxed{210}$$

Step 3. $84 + 126 = 210$, so $\binom{9}{3} + \binom{9}{4} = \binom{10}{4}$ — Pascal's rule. In the triangle, two neighbouring entries of row 9 add to the entry sitting between them in row 10. The reason is a counting one: a group of 4 from 10 people either contains the tenth person (choose 3 more from 9) or does not (choose all 4 from 9).

5. Expand $(2x - 1)^4$.

Step 1. Row 4 is 1, 4, 6, 4, 1, and here $a = 2x$, $b = -1$, so the terms are $\binom{4}{k}(2x)^{4-k}(-1)^k$.

Step 2. Term by term: $(2x)^4 = 16x^4$; $4(2x)^3(-1) = -32x^3$; $6(2x)^2(1) = 24x^2$; $4(2x)(-1) = -8x$; $(-1)^4 = 1$.

Step 3. The signs alternate because b is negative, and the powers of 2 shrink as the power of x does: $16x^4 - 32x^3 + 24x^2 - 8x + 1$. Check at $x = 1$: the expansion gives 1, and $(2 - 1)^4 = 1$ ✓

$$\boxed{16x^4 - 32x^3 + 24x^2 - 8x + 1}$$

6. Exactly 4 heads in 6 flips of a fair coin.

Step 1 (part a). An outcome is fixed by saying *which* flips were heads, so the number of outcomes with exactly 4 heads is $\binom{6}{4} = \frac{6 \cdot 5}{2} = 15$.

$$\boxed{15}$$

Step 2 (part b). All $2^6 = 64$ sequences are equally likely, so the probability is $\frac{15}{64}$.

Step 3. Note where the coefficient came from: 15 is the entry in row 6 of Pascal's triangle, and the whole row sums to 64. That is why the probabilities of 0, 1, ..., 6 heads add to

1.

$$\boxed{\frac{15}{64}}$$

7. Coefficient of x^5 in $(x + 3)^8$.

Step 1. The general term is $\binom{8}{k}x^k3^{8-k}$, so the x^5 term is the one with $k = 5$.

Step 2. $\binom{8}{5} = \binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{6} = 56$, and the remaining factor is $3^3 = 27$.

Step 3. The coefficient is $56 \cdot 27 = 1512$. Answering 56 is the standard slip: the binomial coefficient counts the terms, but the constant raised to its power belongs to the coefficient too.

$$\boxed{1512}$$

8. Exactly 2 red from 5 red and 7 blue, drawing 3.

Step 1 (part a). Choose which reds and which blue independently: $\binom{5}{2} = 10$ and $\binom{7}{1} = 7$, giving $10 \cdot 7 = 70$ favourable draws. $\boxed{10 \cdot 7}$

Step 2 (part b). Total draws of 3 from 12 marbles: $\binom{12}{3} = \frac{12 \cdot 11 \cdot 10}{6} = 220$. $\boxed{220}$

Step 3 (part c). $\frac{70}{220} = \frac{7}{22}$. Note the shape: because the marbles are drawn without replacement, the coefficients multiply on top and divide by a single coefficient below — this is not the $\binom{n}{k} p^k q^{n-k}$ formula, which needs independent trials. $\boxed{\frac{7}{22}}$

9. Exactly 3 made free throws out of 5, with $p = 0.7$.

Step 1. The throws are independent with a fixed success probability, so this is the binomial formula $\binom{5}{3} p^3 (1-p)^2$ with $p = \frac{7}{10}$.

Step 2. $\binom{5}{3} = 10$; one particular sequence of three makes and two misses has probability $\left(\frac{7}{10}\right)^3 \left(\frac{3}{10}\right)^2 = \frac{343}{1000} \cdot \frac{9}{100} = \frac{3087}{100000}$.

Step 3. There are 10 such sequences, so multiply: $10 \cdot \frac{3087}{100000} = \frac{3087}{10000} = 0.3087$. The coefficient counts the arrangements; the powers price one arrangement. $\boxed{\frac{3087}{10000}}$

10. Constant term of $(2x + \frac{1}{x})^6$.

Step 1. General term: $\binom{6}{k} (2x)^{6-k} \left(\frac{1}{x}\right)^k = \binom{6}{k} 2^{6-k} x^{6-2k}$.

Step 2. The term is constant when $6 - 2k = 0$, that is $k = 3$.

Step 3. Evaluate at $k = 3$: $\binom{6}{3} = 20$ and $2^3 = 8$, so the constant term is $20 \cdot 8 = 160$. $\boxed{160}$

11. Error analysis: Jo's $\frac{2}{5}$.

Step 1 (part a). Outcomes with exactly 2 heads in 5 flips: $\binom{5}{2} = \frac{5 \cdot 4}{2} = 10$. $\boxed{10}$

Step 2 (part b). There are $2^5 = 32$ equally likely sequences, so the probability is $\frac{10}{32} = \frac{5}{16}$, about 0.3125 — not 0.4.

Step 3 (part c). Jo compared a count of *flips* with a count of flips, but a probability compares a count of *outcomes* with a count of outcomes. The fraction $\frac{2}{5}$ answers “what share of the flips are heads”, which is a different question; the binomial coefficient $\binom{5}{2}$ is what converts “2 of the 5 flips” into the number of sequences that achieve it.

Manual review (part c): accept any sentence saying that Jo used the ratio of heads to flips instead of the ratio of favourable outcomes to total outcomes, or that the arrangement count

$\binom{5}{2}$ was omitted. $\boxed{\frac{5}{16}}$

12. Synthesis: 8 components, each defective with probability $\frac{1}{4}$.

Step 1 (part a). With $p = \frac{1}{4}$ and $q = \frac{3}{4}$, the expansion of $(q + p)^8$ has terms $\binom{8}{k} q^{8-k} p^k$, and k counts the defective components. Exactly 2 defective is the $k = 2$ term, $\binom{8}{2} \left(\frac{3}{4}\right)^6 \left(\frac{1}{4}\right)^2$, whose binomial coefficient is $\binom{8}{2} = 28$. 28

Step 2 (part b). $\left(\frac{1}{4}\right)^2 = \frac{1}{16}$ and $\left(\frac{3}{4}\right)^6 = \frac{729}{4096}$, so the term is $28 \cdot \frac{1}{16} \cdot \frac{729}{4096} = \frac{5103}{16384}$, about 0.3115. $\frac{5103}{16384}$

Step 3 (part c). Expanding $(q + p)^8$ means choosing q or p from each of the eight factors, one factor per component. A product with exactly two p 's corresponds to one specific set of two defective components, and $\binom{8}{2}$ is precisely the number of ways to choose that set — so the binomial coefficient is not an algebraic accident, it is the count of which components failed. Since $q + p = 1$, the whole expansion sums to 1, which is why the nine terms form a complete probability distribution.

Manual review (part c): accept any explanation linking the choice of p from k of the 8 factors to the choice of which k components are defective.